Q] Verify Rank nulity theorem for the linear transformation defined by $T: \mathbb{R}^3 \to \mathbb{R}^3$ — codomain
$T(x, y, z) = (x, 0, x - y + z)$ domain

$\Rightarrow$ matrix of L.T T w.r.t standard basis $\{(1,0,0), (0,1,0), (0,0,1)\}$ is

$$A = \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \end{array}$$

Range (T) = col(A) = span columns of A

$= \text{span}\{(1,0,1), (0,0,-1), \cancel{(0,0,1)}\}$ dependent

Basis of (Range (T)) = Basis of col(A).

$= \{(1,0,1), (0,0,-1)\}$.

$\text{Rank}(T) = \dim(\text{range}(T)) = \underline{2}$.

$\text{Nul}(T) = \{(x, y, z) \mid T(x, y, z) = 0 = (0,0,0)\}$

$\{(x, y, z) \mid (x, 0, x - y + z) = (0,0,0)\}$

$\{x, y, z \mid x = 0, 0 = 0, x - y + z = 0\}$

$$\left\{(x, y, z) \mid \begin{array}{l} x=0,\ 0-y+z \\ y=z \end{array}\right\}$$

$\cancel{(Null\ T) = \{(x, y, z)\}}$

$Nul(T) = \{(0, y, y) \mid y \in \mathbb{R}\}$

$= \{y(0, 1, 1) \mid y \in \mathbb{R}\}$

$= \text{span}\{(0, 1, 1)\}$

Basis of $Nul(T) = \{(0, 1, 1)\}$

$Nullity(T) = \dim(Nul(T)) = 1$

$\therefore Rank(T) + Nullity(T)$

$2 + 1 = \underline{3} = \dim(\mathbb{R}^3)$

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② find range space, null space Rank & nullity of L.T of T, $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x+y,\ x-y,\ x+2z)$.

$\Rightarrow$ matrix of L.T T w.r.t standard basis $\{(1,0,0), (0,1,0), (0,0,1)\}$

independent

$$A = \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} \boxed{1} & 1 & 0 \\ 1 & \boxed{-1} & 0 \\ 1 & 0 & \boxed{2} \end{bmatrix} \end{array}$$

$$\text{Range}(T) = \text{col}(A)$$
$$= \text{span of col}(A)$$
$$= \text{span}\{(1,1,0), (1,-1,0), (0,0,2)\}$$

$R_3 \to R_3 - R_1$
$R_2 \to R_2 - R_1$

$$A \sim \begin{bmatrix} \boxed{1} & 1 & 0 \\ 0 & \boxed{-2} & 0 \\ 0 & -1 & 2 \end{bmatrix}$$

$R_3 \to 2R_3 - R_2$

$$\sim \begin{bmatrix} \boxed{1} & 1 & 0 \\ 0 & \boxed{-2} & 0 \\ 0 & 0 & \boxed{4} \end{bmatrix}$$

Basis of Range(T) = Basis of col(A)
$= \{(1,1,0), (1,-1,0), (0,0,2)\}$

$\text{Rank}(T) = \dim(\text{Range}(T)) = \underline{\underline{3}}$.

$$\text{Null}(T) = \{(x,y,z) \mid T(x,y,z) = 0 = (0,0,0)\}$$

$$\{(x,y,z) \mid (x+y, x-y, x+2z) = (0,0,0)\}.$$

$$\{(x,y,z) \mid x+y=0,\ x-y=0,\ x+2z=0\}$$

$x = -y$ $\quad x + x = 0$ $\quad \underline{\underline{z=0}}$.
$y = 0$ $\quad 2x = 0$
$\underline{x=0}$.

$Null(T) = \{(0,0,0)\}$

Basis of $Null(T) = \cancel{\{(0,0,0)\}}$ $\{\}$

$Nullity(T) = \dim(Null(T)) = 0$

$= Rank(T) + Nulity(T)$

$= 3 + 0 = 3 = \dim(\mathbb{R}^3)$

Matrix of the linear transformation relative to the basis:

Let $T: V \to W$ be a linear transformation suppose

$B = \{u_1, u_2, \ldots, u_n\}$ is a basis of V &
$B' = \{v_1, v_2, \ldots, v_n\}$ is a basis of W

Then the vectors $T(u_1), T(u_2), \ldots, T(u_n)$ in W, can be written as a linear ~~transformation~~ combination in the Basis that is...

Defn:- The transpose of the matrix of coefficients in the above eqns, denoted by $[T]_{B,B'}$ is called the matrix representation of T relative to the bases B and B'.

i.e., $= [T]_{B,B'} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$

$T(u_i) = a_{11}v_1 + a_{21}v_2 \ldots + a_{m1}v_n$.

using the coordinate vector notation, the matrix representation of T relative to the basis B & B' may be written in the form,

$$[T]_{B,B'} = [[T(u_1)]_{B'}, [T(u_2)]_{B'}, \ldots, [T(u_n)]_{B'}]$$

To find matrix of L.T.

① find images of $u_1, u_2, \ldots u_n$
$T(u_1), T(u_2), \ldots T(u_n)$.

② coordinate vectors of $T(u_1), T(u_2), \ldots T(u_n)$ relative to basis $B' = \{v_1, v_2, \ldots v_n\}$

① find the matrix transformation of $T: \mathbb{R}^2 \to \mathbb{R}^2$ defined, $T(x,y) = (x, -y)$.

$B_1 = \{(1,1), (1,0)\}$ } given
$B_2 = \{(2,3), (4,5)\}$

$\Rightarrow$ Let $u_1 = (1,1)$, $u_2 = (1,0)$ $\longrightarrow$ domain
$v_1 = (2,3)$, $v_2 = (4,5)$ $\longrightarrow$ codomain

$T(u_1) = T(1,1) = (1,-1)$ //

$= a_{11} v_1 + a_{21} v_2$

$T(u_1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = a_{11} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + a_{21} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$2a_{11} + 4a_{21} = 1$
$3a_{11} + 5a_{21} = -1$

$a_{11} = -4.5$
$a_{21} = 2.5 = 5/2$

$T(u_2) = T(1,0) = (1,0)$ //

$= a_{12} v_1 + a_{22} v_2$

$T(u_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_{22} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + a_{22} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$$2a_{12} + 4a_{22} = 1$$
$$3a_{12} + 5a_{22} = 0$$

$$a_{12} = -2.5$$
$$a_{22} = 1.5$$

matrix of L.T is,

$$[T]_{B_1,B_2} = \begin{bmatrix} -4.5 & -2.5 \\ 2.5 & 1.5 \end{bmatrix}$$

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(2) find the matrix transformation $T: V_2 \to V_3$ defined by $T(x,y) = (x+y,\ x,\ 3x-y)$

$B_1 = \{(1,1), (3,1)\}$

$B_2 = \{(1,1,1), (1,1,0), (1,0,0)\}$

$\Rightarrow u_1 = (1,1) \quad u_2 = (3,1)$

$v_1 = (1,1,1) \quad v_2 = (1,1,0) \quad v_3 = (1,0,0)$

$T(u_1) = T(1,1) = (2, 1, 2)$

$= a_1 v_1 + a_2 v_2 + a_3 v_3$

$$\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 + a_2 + a_3 = 2$$
$$a_1 + a_2 = 1$$
$$\boxed{a_1 = 2}$$

$$a_1 + a_2 = 1$$
$$2 + a_2 = 1$$
$$\boxed{a_2 = -1}$$

$$a_3 + 2 - 1 = 2$$
$$\boxed{a_3 = 1}$$

$$T(u_2) = T(3,1) = (4, 3, 8)$$
$$= b_1 v_1 + b_2 v_2 + b_3 v_3$$

$$\begin{bmatrix} 4 \\ 3 \\ 8 \end{bmatrix} = b_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + b_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$b_1 + b_2 + b_3 = 4$$
$$b_1 + b_2 = 3$$
$$\boxed{b_1 = 8}$$

$$8 + b_2 = 3$$
$$b_2 = 3 - 8$$
$$\boxed{b_2 = -5}$$

$8 - 5 + b_3 = 4$

$b_3 \quad 4 - 3$

$\boxed{b_3 = 1}$

$$[T]_{B_1, B_2} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ -1 & -5 \\ 1 & 1 \end{bmatrix}$$

(4) find the M.T from $T: \mathbb{R}^3 \to \mathbb{R}^3$ $T(x, y, z) = (x - y + 2z, 3x + y)$

$B_1 = \{(1,1,1), (1,2,3), (1,0,0)\}$

$B_2 = \{(1,1)(1,-1)\}$

Dot product / scalar product:

Let $u, v \in \mathbb{R}^n$, then the dot product of u & v is denoted by $u \cdot v$, defined as,

$$u \cdot v = u^T v.$$

If $u = (u_1, \dots, u_n)$ $\quad v = (v_1, v_2 \dots, v_n)$

then $u \cdot v = u^T v = [u_1 \; u_2 \dots u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_3 \end{bmatrix}$

$= u_1 v_1 + u_2 v_2 + \dots + u_3 v_3$

$= \underline{v \cdot u}$

Ex:- Let $u = (1, -2, 3, 5)$ & $v = (1, -1, 0, 5)$

$u \cdot v = (1 \times 1) + (-2 \times -1) + (3 \times 0) + (5 \times 5)$

$= 1 + 2 + 25$

$= \underline{28} = v \cdot u$

$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2}$

$= \sqrt{1 + 1 + 0 + 25}$

$= \underline{\sqrt{27}}$

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orthogonal vectors: Two vectors u and v are said to be orthogonal or perpendicular vector if dot product of u & v i.e. $\boxed{u \cdot v = 0}$ or the angle betn two vectors should be equal to $90°$.

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orthogonal set.

A set $\{u_1, u_2, \ldots, u_n\}$ of vectors is said to be orthogonal if $u_i \cdot u_j = 0$ for all i, j with $i \neq j$.

Note:- Let $\{u_1, u_2, \ldots, u_n\}$ be an orthogonal set of non-zero vectors then,

$\{u_1, u_2, u_3, \ldots, u_n\}$ is linearly independent.

eg:- Let $u_1 = (3,1,1)$ $u_2 = (-1,2,1)$ $u_3 = (-1/2, -2, 7/2)$

$u_1 \cdot u_2 = (3 \times -1) + (1 \times 2) + (1 \times 1) = -3 + 2 + 1 = \underline{\underline{0}}$

$u_1 \cdot u_3 = (3 \times -1/2) + (1 \times -2) + (1 \times 7/2) = -3/2 - 2 + 7/2 = 0$

$u_2 \cdot u_3 = 1/2 - 4 + 7/2 = 0$

<u>orthonormal set</u>: An orthogonal set $\{v_1, v_2, \ldots, v_n\}$ of vectors is orthonormal if $\|v_i\| = 1$, for all i.

Ex:- Let $u_1 = (3,1,1)$, $u_2 = (-1,2,1)$, $u_3 = (-1/2, -2, 7/2)$

$$\|u_1\| = \sqrt{3^2 + 1^2 + 1^2} = \sqrt{9+1+1} = \underline{\underline{\sqrt{11}}}.$$

$$\frac{u_1}{\|u_1\|} = \underbrace{\frac{1}{\sqrt{11}}(3,1,1)}_{\text{unit vector}} = v_1$$

$$\|u_2\| = \sqrt{1+4+1} = \underline{\underline{\sqrt{6}}}.$$

$$\frac{u_1}{\|u_2\|} = \frac{1}{\sqrt{6}}(-1,2,1) = v_2$$

$$\|u_3\| = \sqrt{(-1/2)^2 + (-2)^2 + (7/2)^2}$$

$$\sqrt{0.25 + 4 + 12.25}$$

$$\sqrt{16.5} = \underline{\underline{4.06}}$$

$$\frac{u_3}{\|u_3\|} = \frac{1}{4.06}(-1/2, -2, 7/2) = v_3$$

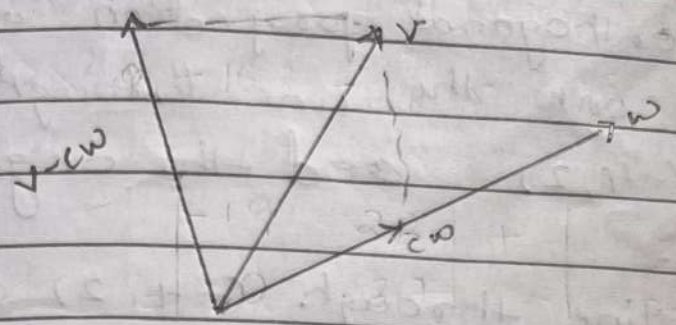
$\{v_1, v_2, v_3\}$ is a orthonormal set.

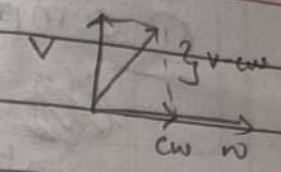
orthogonal projection of vector.

Let w be a vector in R^n. Suppose v is an another vector. we seek the "projection of v along w," which, as indicated in figure, will be the multiple cw of w such that $v - cw$ is orthogonal to w. This means.

$$(v - cw) \cdot w = 0 \text{ (or)}$$

$$c = \frac{v \cdot w}{w \cdot w}$$





Then the vector $cw - \hat{v} = \text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$ is called orthogonal projection of v onto w, & the vector $v - \hat{v}$ is called the component of v orthogonal to w.

If k is any non-zero scalar if w is replaced by kw in the defn of $\hat{v}$, then the orthogonal projection of v onto kw is exactly the same as the orthogonal projection of v onto w. hence this projection is determined by the subspace W spanned by w (that is the line through w & 0). Sometimes $\hat{v}$ is denoted by $\text{proj}_w v$ is called the orthogonal projection of v onto w.

i.e, $\hat{v} = \text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$